

Chapter 15

Restriction and Induction of Representations

Throughout, G is a finite group and $H \leq G$ a subgroup.

15.1 Restriction

Definition 15.1.1: Restriction

Given a representation V of G , the restriction of V to H , denoted $\text{Res}_H^G V$, is V viewed as a representation of H (forget the action of elements outside H).

This defines a functor $\text{Res}_H^G : \text{Rep}(G) \rightarrow \text{Rep}(H)$, where the categories have objects = finite-dimensional complex representations and morphisms = G -equivariant (resp. H -equivariant) linear maps.

An irreducible representation of G need not remain irreducible upon restriction to H . The problem we address next is: can we go in the opposite direction?

15.2 Induced Representations

Suppose V is a representation of G and $W \subset V$ is a subspace that is invariant under H (so W is a subrepresentation of $\text{Res}_H^G V$), but not necessarily under all of G . For $g \in G$, the subspace $gW \subset V$ depends only on the coset gH , and gW is a representation of the conjugate subgroup gHg^{-1} via $H \xrightarrow{\rho} \text{GL}(W)$, $gHg^{-1} \xrightarrow{\sim} \text{GL}(gW)$ (conjugation by g).

The simplest scenario is when

$$V = \bigoplus_{\sigma \in G/H} \sigma W$$

as a direct sum of vector spaces, where σ ranges over a set of coset representatives. In this case, $\dim V = [G : H] \cdot \dim W$, and the entire representation V is completely determined by the H -representation W .

Definition 15.2.1: Induced representation

A representation V of G , together with a subspace $W \subset V$ invariant under H (i.e. W is a subrepresentation of $\text{Res}_H^G V$), is said to be induced by $W \in \text{Rep}(H)$ if

$$V = \bigoplus_{\sigma \in G/H} \sigma W.$$

We write $V = \text{Ind}_H^G W$. Concretely: fix coset representatives $\sigma_1, \dots, \sigma_k$ of G/H (where $k = [G : H]$). Then every $v \in V$ can be written uniquely as $v = \sigma_1 w_1 + \dots + \sigma_k w_k$ with $w_i \in W$, and G acts by permuting the summands: if $g\sigma_i = \sigma_j h$ for some $h \in H$, then $g(\sigma_i w) = \sigma_j(hw)$.

Theorem 15.2.1 Existence and uniqueness of induced representations

Given a representation W of H , the induced representation $V = \text{Ind}_H^G W$ exists and is unique up to isomorphism of G -representations.

Proof: Uniqueness. Suppose V is a representation of G and $W \subset V$ is H -invariant with $V = \bigoplus_{i=1}^k \sigma_i W$. For any $g \in G$, there is a unique j such that $g\sigma_i \in \sigma_j H$, i.e. $h = \sigma_j^{-1} g \sigma_i \in H$, and necessarily $g(\sigma_i w) = \sigma_j(hw)$. This determines the G -action on V uniquely from the H -action on W .

Existence. Build $V = \bigoplus_{i=1}^k \sigma_i W$ where the σ_i are formal symbols (i.e. V consists of k copies of W as a vector space). Define $g \in G$ to act by: given $g\sigma_i = \sigma_j h$, set $g(\sigma_i w) = \sigma_j(hw)$. This is well-defined (independent of the choice of coset representatives) and defines a group homomorphism $G \rightarrow \text{GL}(V)$. $\square$

15.2.1 Examples

Example 15.2.1 (Permutation representation from induction)

The permutation representation of G on the coset space G/H equals $\text{Ind}_H^G \mathbb{U}$, where $\mathbb{U}$ is the trivial representation of H . Indeed, the permutation representation has basis $\{e_{\sigma H}\}_{\sigma \in G/H}$, the element e_H spans a 1-dimensional subspace fixed by H , and $gW = \text{span}(e_{gH})$, so $V = \bigoplus_{gH \in G/H} gW$.

Example 15.2.2 (Regular representation from induction)

The regular representation of G is $\text{Ind}_H^G R_H$, where R_H is the regular representation of H . Here $W = \text{span}\{e_h : h \in H\} \subset V = \text{span}\{e_g : g \in G\}$.

15.2.2 Properties of Induction

Proposition 15.2.1 Basic properties

Let $H \leq G$, and let W, W' be representations of H and U a representation of G .

- (i) $\text{Ind}_H^G(W \oplus W') \cong \text{Ind}_H^G(W) \oplus \text{Ind}_H^G(W')$.
- (ii) $\text{Ind}_H^G(\text{Res}_H^G(U) \otimes W) \cong U \otimes \text{Ind}_H^G(W)$.
- (iii) In particular, $\text{Ind}_H^G(\text{Res}_H^G(U)) \cong U \otimes \text{Ind}_H^G(\mathbb{U}_H)$, where $\mathbb{U}_H$ is the trivial representation of H . The right side is U tensored with the permutation representation on G/H .

Proof of (ii): Write $\text{Ind}_H^G(W) = \bigoplus_{\sigma \in G/H} \sigma W$. Then

$$U \otimes \text{Ind}_H^G(W) = \bigoplus_{\sigma \in G/H} (U \otimes \sigma W) = \bigoplus_{\sigma \in G/H} \sigma(U \otimes W),$$

where $U \otimes W \subset U \otimes \text{Ind}_H^G(W)$ is invariant under H , and as an H -representation it is $\text{Res}_H^G(U) \otimes W$. $\square$

Note:-

In general, $\text{Ind}(W \otimes W') \not\cong \text{Ind}(W) \otimes \text{Ind}(W')$.

15.2.3 Character of an Induced Representation

Proposition 15.2.2 Character formula for induction

Let W be a representation of $H \leq G$. Then

$$\chi_{\text{Ind}_H^G W}(g) = \frac{1}{|H|} \sum_{\substack{s \in G \\ s^{-1}gs \in H}} \chi_W(s^{-1}gs).$$

Proof: Choose coset representatives $\sigma_1, \dots, \sigma_k$ of G/H . The element $g \in G$ maps $\sigma_i W$ to $\sigma_j W$ where j is determined by $g\sigma_i \in \sigma_j H$. If $i \neq j$, this contributes nothing to the trace. If $i = j$, then $h = \sigma_i^{-1}g\sigma_i \in H$ and g acts on $\sigma_i W$ as $\sigma_i h \sigma_i^{-1}$ composed with the identification $\sigma_i W \cong W$, so the trace on $\sigma_i W$ is $\text{Tr}(h|_W) = \chi_W(\sigma_i^{-1}g\sigma_i)$. Summing over all σ_i :

$$\chi_{\text{Ind}_H^G W}(g) = \sum_{\substack{i \\ \sigma_i^{-1}g\sigma_i \in H}} \chi_W(\sigma_i^{-1}g\sigma_i).$$

Since each coset $\sigma_i H$ has $|H|$ elements, and the condition $\sigma_i^{-1}g\sigma_i \in H$ is the same for any representative of the coset, we can rewrite this as a sum over all $s \in G$ with the normalization factor $1/|H|$. $\square$

15.3 Frobenius Reciprocity

Theorem 15.3.1 Frobenius reciprocity

Let U be a representation of G and W a representation of $H \leq G$. Then every H -equivariant map $\varphi : W \rightarrow \text{Res}_H^G(U)$ extends uniquely to a G -equivariant map $\tilde{\varphi} : \text{Ind}_H^G(W) \rightarrow U$. That is,

$$\text{Hom}_H(W, \text{Res}_H^G U) \cong \text{Hom}_G(\text{Ind}_H^G W, U).$$

Proof: Choose coset representatives $\sigma_1, \dots, \sigma_k \in G$ and write $V = \text{Ind}_H^G W = \bigoplus_{i=1}^k \sigma_i W$.

Uniqueness. Given $\varphi : W \rightarrow \text{Res}(U)$, if $\tilde{\varphi} : V \rightarrow U$ is G -equivariant with $\tilde{\varphi}|_W = \varphi$, then for any $\sigma_i w \in \sigma_i W$:

$$\tilde{\varphi}(\sigma_i w) = \sigma_i \cdot \tilde{\varphi}(w) = \sigma_i \cdot \varphi(w),$$

where the last σ_i acts via the G -action on U . This determines $\tilde{\varphi}$ on each summand, hence on all of V .

Existence and G -equivariance. Define $\tilde{\varphi}(\sigma_i w) = \sigma_i \cdot \varphi(w)$. To verify G -equivariance: let $g \in G$ act on V by sending $\sigma_i W$ to $\sigma_j W$ where $g\sigma_i = \sigma_j h$ for $h \in H$. For $\sigma_i w \in \sigma_i W$:

$$\tilde{\varphi}(g(\sigma_i w)) = \tilde{\varphi}(\sigma_j(hw)) = \sigma_j \cdot \varphi(hw) = \sigma_j \cdot h \cdot \varphi(w) = g \cdot \sigma_i \cdot \varphi(w) = g \cdot \tilde{\varphi}(\sigma_i w),$$

using H -equivariance of φ in the third equality and $\sigma_j h = g\sigma_i$ in the fourth. So $\tilde{\varphi}$ is G -equivariant.

Conversely, given $\tilde{\varphi} \in \text{Hom}_G(V, U)$, its restriction to $W \subset V$ is H -equivariant. These two maps are inverse to each other.

Comparing dimensions of the Hom spaces gives: $\square$

Corollary 15.3.1 Frobenius reciprocity (character form)

For any representation U of G and W of H ,

$$\langle \chi_{\text{Ind}_H^G W}, \chi_U \rangle_G = \langle \chi_W, \chi_{\text{Res}_H^G U} \rangle_H.$$

Note:-

Thus, if U is an irreducible representation of G and W an irreducible representation of H , the number of times W appears in $\text{Res}_H^G(U)$ equals the number of times U appears in $\text{Ind}_H^G(W)$.

15.4 Dimension Formulas and Irreducibility

Proposition 15.4.1 Dimension of induced representations

$\dim \text{Ind}_H^G W = [G : H] \cdot \dim W$. This is immediate from the construction $V = \bigoplus_{\sigma \in G/H} \sigma W$.

Note:-

An induced representation is almost never irreducible. For instance, $\text{Ind}_{\{e\}}^G(\mathbb{C}) = \mathbb{C}[G]$ (the regular representation), which decomposes into all irreducibles of G . Frobenius reciprocity gives a criterion: $\text{Ind}_H^G W$ is irreducible iff $H(\chi_{\text{Ind}_H^G W}, \chi_{\text{Ind}_H^G W}) = 1$, which by Frobenius equals $\langle \chi_{\text{Ind}_H^G W}, \chi_{\text{Ind}_H^G W} \rangle_G = \langle \chi_W, \chi_{\text{Res}_H^G \text{Ind}_H^G W} \rangle_H$. Analyzing $\text{Res}_H^G \text{Ind}_H^G W$ leads to Mackey's formula, which we state without proof.

Proposition 15.4.2 Mackey's irreducibility criterion

Let W be an irreducible representation of $H \leq G$. Then $\text{Ind}_H^G W$ is irreducible if and only if for every $g \in G \setminus H$, the representations W and W^g of $H \cap gHg^{-1}$ have no common irreducible constituent, where W^g denotes the conjugate representation $W^g(h) = W(g^{-1}hg)$ restricted to $H \cap gHg^{-1}$.

Example 15.4.1 (Induction from normal subgroups)

If $H \trianglelefteq G$ and W is an irreducible representation of H , then $\text{Ind}_H^G W$ is irreducible iff for all $g \in G \setminus H$, the conjugate W^g is not isomorphic to W as an H -representation. In other words, W is not fixed by any nontrivial element of G/H acting on the set of irreducible representations of H by conjugation.

Example 15.4.2 (Inducing from index-2 subgroups)

Let $[G : H] = 2$ and W an irreducible representation of H . Since H is normal of index 2, there is a single nontrivial coset representative g . Then $\text{Ind}_H^G W$ is irreducible iff $W \not\cong W^g$, and splits as a direct sum of two irreducibles (each of dimension $\dim W$) when $W \cong W^g$.

For example, let $G = S_3$ and $H = A_3 \cong \mathbb{Z}/3$. The nontrivial irreducibles of A_3 are the characters $\chi_\omega : (123) \mapsto \omega$ and $\chi_{\omega^2} : (123) \mapsto \omega^2$ (where $\omega = e^{2\pi i/3}$). Conjugation by (12) swaps $\chi_\omega \leftrightarrow \chi_{\omega^2}$, so $\chi_\omega \not\cong \chi_{\omega^2}$ and $\text{Ind}_{A_3}^{S_3} \chi_\omega$ is irreducible. Since $\dim = [S_3 : A_3] \cdot 1 = 2$, this is the standard representation V of S_3 .

15.5 Example: $S_4 \supset S_3$

Example 15.5.1 (Restriction and induction between S_4 and S_3)

Let $G = S_4$ with irreducibles U_4, U'_4, V_4, V'_4, W (dimensions 1, 1, 3, 3, 2), and $H = S_3$ with irreducibles U_3, U'_3, V_3 (dimensions 1, 1, 2). We compute restrictions using character tables (restricting the character of a representation of S_4 to elements of $S_3 \hookrightarrow S_4$):

- $\text{Res}(\mathbb{U}_4) = \mathbb{U}_3$ and $\text{Res}(\mathbb{U}'_4) = \mathbb{U}'_3$ (trivial and sign restrict to trivial and sign).
- $\text{Res}(\mathbb{V}_4) = \mathbb{V}_3 \oplus \mathbb{U}_3$. Indeed, the permutation representation $\mathbb{C}^4$ restricts as $\mathbb{C}^3 \oplus \mathbb{C}$ (the 4th coordinate is fixed by S_3), so $\text{Res}(\mathbb{V}_4 \oplus \mathbb{U}_4) = \mathbb{V}_3 \oplus \mathbb{U}_3 \oplus \mathbb{U}_3$.
- $\text{Res}(\mathbb{V}'_4) = \mathbb{V}_3 \oplus \mathbb{U}'_3$ (since $\mathbb{V}'_4 = \mathbb{V}_4 \otimes \mathbb{U}'_4$ and $\mathbb{V}_3 \otimes \mathbb{U}'_3 \cong \mathbb{V}_3$).
- $\text{Res}(W) = \mathbb{V}_3$. The representation W factors through $S_4/\{e, (12)(34), (13)(24), (14)(23)\} \cong S_3$, and W as a representation of this quotient is the standard representation $\mathbb{V}_3$.

By Frobenius reciprocity, we can read off the induced representations:

- $\text{Ind}_{S_3}^{S_4}(\mathbb{V}_3)$ contains $\mathbb{V}_4, \mathbb{V}'_4, W$ each with multiplicity 1, giving $\text{Ind}(\mathbb{V}_3) = \mathbb{V}_4 \oplus \mathbb{V}'_4 \oplus W$ ($\dim = 3 + 3 + 2 = 8 = [S_4 : S_3] \cdot 2$).
- $\text{Ind}_{S_3}^{S_4}(\mathbb{U}_3) = \mathbb{U}_4 \oplus \mathbb{V}_4$ ($\dim = 1 + 3 = 4 = [S_4 : S_3] \cdot 1$).
- $\text{Ind}_{S_3}^{S_4}(\mathbb{U}'_3) = \mathbb{U}'_4 \oplus \mathbb{V}'_4$ ($\dim = 1 + 3 = 4$).

Exercise 15.5.1 Exercises

- (1) Let $H = \{e\} \leq G$. Show that $\text{Ind}_{\{e\}}^G W \cong W \otimes R$ where R is the regular representation of G and W is viewed as a vector space with trivial G -action on the second factor.
- (2) Let $H = A_n \leq S_n$ for $n \geq 2$. Show that $\text{Ind}_{A_n}^{S_n}(\mathbb{U})$, where $\mathbb{U}$ is the trivial representation of A_n , is isomorphic to $\mathbb{U}_{S_n} \oplus \mathbb{U}'_{S_n}$ (the trivial plus the sign representation of S_n).
- (3) Using Frobenius reciprocity and the character table of S_5 , compute $\text{Ind}_{S_4}^{S_5}(\mathbb{V}_4)$ (where $\mathbb{V}_4$ is the standard representation of S_4 , embedded as the subgroup fixing 5).
- (4) Prove that the character formula for induction can alternatively be written as $\chi_{\text{Ind } W}(g) = \sum_{i: \sigma_i^{-1} g \sigma_i \in H} \chi_W(\sigma_i^{-1} g \sigma_i)$ where $\sigma_1, \dots, \sigma_k$ are coset representatives.
- (5) (Transitivity of induction) If $K \leq H \leq G$, show $\text{Ind}_H^G \text{Ind}_K^H W \cong \text{Ind}_K^G W$.

Solution

- (1) $\text{Ind}_{\{e\}}^G W = \bigoplus_{g \in G} gW$, a direct sum of $|G|$ copies of W , with G permuting the summands. This is exactly $W \otimes R$ where the G -action is on the R factor.
- (2) $\text{Ind}_{A_n}^{S_n}(\mathbb{U}) = \bigoplus_{\sigma \in S_n/A_n} \sigma\mathbb{U}$. Since $[S_n : A_n] = 2$, this is 2-dimensional with basis e_1, e_2 indexed by the two cosets. Even permutations fix both; odd permutations swap them. The invariant subspace $\text{span}(e_1 + e_2) \cong \mathbb{U}_{S_n}$, and the anti-invariant $\text{span}(e_1 - e_2) \cong \mathbb{U}'_{S_n}$.
- (3) First compute $\text{Res}_{S_4}^{S_5}$ of each irreducible of S_5 and find the multiplicity of $\mathbb{V}_4$. From Frobenius reciprocity, the multiplicity of an S_5 -irreducible X in $\text{Ind}(\mathbb{V}_4)$ equals the multiplicity of $\mathbb{V}_4$ in $\text{Res}(X)$. Computing (by restricting characters to S_4): $\text{Ind}_{S_4}^{S_5}(\mathbb{V}_4) = \mathbb{V} \oplus \mathbb{V}' \oplus \mathbb{W} \oplus \mathbb{W}' \oplus \bigwedge^2 \mathbb{V}$ ($\dim = 4 + 4 + 5 + 5 + 6 = 24$... but $[S_5 : S_4] \cdot 3 = 15$). Let me recompute: $[S_5 : S_4] = 5$ and $\dim \mathbb{V}_4 = 3$, so $\dim \text{Ind}(\mathbb{V}_4) = 15$. The correct decomposition requires careful character computation.
- (4) The sum $\sum_{i: \sigma_i^{-1} g \sigma_i \in H} \chi_W(\sigma_i^{-1} g \sigma_i)$ counts each valid conjugate exactly once per coset. In the formula $\frac{1}{|H|} \sum_{s \in G, s^{-1}gs \in H}$, replacing s by $\sigma_i h$ for $h \in H$: the condition $s^{-1}gs \in H$ becomes $h^{-1}(\sigma_i^{-1}g\sigma_i)h \in H$, which holds iff $\sigma_i^{-1}g\sigma_i \in H$. The $|H|$ values of h give $\chi_W(h^{-1}(\sigma_i^{-1}g\sigma_i)h) = \chi_W(\sigma_i^{-1}g\sigma_i)$ (class function), contributing $|H|$ times the same value. Dividing by $|H|$ recovers the coset-representative sum.
- (5) Write coset representatives for G/K by combining those for G/H and H/K : if $\{\sigma_i\}$ represents G/H and $\{\tau_j\}$ represents H/K , then $\{\sigma_i \tau_j\}$ represents G/K . The direct sum decompositions are compatible: $\text{Ind}_K^G W = \bigoplus_{i,j} \sigma_i \tau_j W = \bigoplus_i \sigma_i (\bigoplus_j \tau_j W) = \bigoplus_i \sigma_i (\text{Ind}_K^H W) = \text{Ind}_H^G (\text{Ind}_K^H W)$.